DIFFERENTIAL COHOMOLOGY SEMINAR 11

TALK BY MATTHIAS FRERICHS

1. Differential Cohomology in a Cohesive ∞-topos

1.1. **Sheaves on manifolds are cohesive.** We consider the essentially small category of manifolds with corners, denoted $\mathcal{M}\mathrm{fd}$. This category is endowed with the Grothendieck topology generated by surjective submersions with discrete fibers. This is also a dense subsite of the the site of all smooth manifolds with jointly surjective open covers.

Definition 1.1. Let $\mathcal{C}$ be a $(\infty,1)$-category. The category of presheaves on the site of manifolds with values in $\mathcal{C}$ is denoted by $\mathcal{F}\mathrm{un}(\mathcal{M}\mathrm{fd}^{\mathrm{op}},\mathcal{C})$.

We will only consider presentable $(\infty,1)$-categories here, even if [BNV16] treat the more general case too.

Definition 1.2. ([BNV16, Def. 2.3.], Sheaf on Manifolds) A functor $F\in\mathcal{F}\mathrm{un}(\mathcal{M}\mathrm{fd}^{\mathrm{op}},\mathcal{C})$ is a sheaf if for any manifold M and covering $U\to M$ the canoncial map

$$F(M)\to\lim_{\mathbf{\Delta}}F(U^\bullet)$$

is an equivalence. Here $U^\bullet$ is the Čech nerve of the covering and $F(U^\bullet)$ is the simplical object in $\mathcal{F}\mathrm{un}(\mathbf{\Delta}^{\mathrm{op}},\mathcal{C})$ by applying F to the Čech nerve.

Denote the full subcategory of sheaves by $\mathcal{S}\mathrm{hv}_{\mathcal{M}\mathrm{fd}}(\mathcal{C})\subset\mathcal{F}\mathrm{un}(\mathbf{\Delta}^{\mathrm{op}},\mathcal{C})$ which is a reflecive localization given by

$$L\colon\mathcal{F}\mathrm{un}(\mathbf{\Delta}^{\mathrm{op}},\mathcal{C})\rightleftarrows\mathcal{S}\mathrm{hv}_{\mathcal{M}\mathrm{fd}}(\mathcal{C})\colon\text{inclusion}.$$

Remark 1.3. ([BNV16, Def. 2.3.], Homotopy-invariance) A presheaf $F\in\mathcal{F}\mathrm{un}(\mathbf{\Delta}^{\mathrm{op}},\mathcal{C})$ is called homotopy invariant, if for all manifolds $M\in\mathcal{M}\mathrm{fd}$ the canonical map $F(M)\to F(\mathbf{\Delta}^1\times M)$ induced by the projection is an equivalence.

There is an adjunction

$$\mathcal{H}^{\mathrm{pre}}\colon\mathcal{F}\mathrm{un}(\mathbf{\Delta}^{\mathrm{op}},\mathcal{C})\rightleftarrows\mathcal{F}\mathrm{un}^h(\mathbf{\Delta}^{\mathrm{op}},\mathcal{C})\colon\text{inclusion}$$

where $\mathcal{H}^{\mathrm{pre}}$ is called homotopification.

This functor commutes with sheaffication and restricts to sheaves yielding

$$\mathcal{H}\colon\mathcal{S}\mathrm{hv}_{\mathcal{M}\mathrm{fd}}(\mathcal{C})\rightleftarrows\mathcal{S}\mathrm{hv}^h_{\mathcal{M}\mathrm{fd}}(\mathcal{C})\colon\text{inclusion}.$$

By [BNV16, Prop. 2.6.] this functor preserves finite products for the special case that $\mathcal{C}$ is stable or the category of spaces. In our case we will only consider the category of spaces.

The category $\mathcal{S}\mathrm{hv}^h_{\mathcal{M}\mathrm{fd}}(\mathcal{C})$ is equivalent to $\mathcal{C}$ by sheaffication of the constant presheaf and evaluation at the point

$$\mathrm{const}\colon\mathcal{C}\rightleftarrows\mathcal{S}\mathrm{hv}^h_{\mathcal{M}\mathrm{fd}}(\mathcal{C})\colon\mathrm{ev}_*.$$

In conclusion we have the following quadruple of adjoint functors.

Remark 1.4. ([BNV16, Rmk. 2.7.], Coherent Topos) There is an adjoint quadruple of functors

$$(\mathrm{ev}_*\circ\mathcal{H}\dashv\mathrm{const}\dashv\mathrm{ev}_*\dashv\mathcal{G})\colon\mathcal{S}\mathrm{hv}_{\mathcal{M}\mathrm{fd}}(\mathcal{C})\ \begin{tikzcd}[column sep=small]\arrow[r, shift left=3, "\mathrm{ev}_*\circ\mathcal{H}"]\arrow[r, shift left=1, "\mathrm{const}"]\arrow[r, shift right=1, "\mathrm{ev}_*"]\arrow[r, shift right=3, "G"] & \end{tikzcd}\ \mathcal{C}$$

where $G\colon\mathcal{C}\to\mathcal{S}\mathrm{hv}_{\mathcal{M}\mathrm{fd}}(\mathcal{C})$ is defined by $G(F)(M):=\mathrm{const}(F)(*)^{M_{\mathrm{disc}}}=F^{M_{\mathrm{disc}}}$. Here $\mathrm{ev}_*\circ\mathcal{H}$ preserves products ([BNV16, Prop. 2.6.]) and const and G are fully faithful.

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We will only consider cohesive topoi over the ∞-category of spaces.

Definition 1.5. ([Sch13, Rmk. 3.4.2.]) A ∞-topos $\mathcal{X}$ is called cohesive if it admits a quadruple adjoint

$$(\Pi \dashv \mathrm{Disc} \dashv \Gamma \dashv \mathrm{coDisc})\colon \mathcal{X} \underset{\mathrm{coDisc}}{\overset{\Pi}{\underset{\Gamma}{\overset{\longrightarrow}{\underset{\longleftarrow}{\overset{\longleftarrow}{\underset{\longrightarrow}{\mathrm{Disc}}}}}}}} \mathcal{C}$$

such that Π preserves finite products and Disc and coDisc are fully faithfull.

Now we can state the definition of the intrinsic non-abelian cohomology in a cohesive ∞-topos as in [Sch13, Def. 3.6.134.].

Definition 1.6. ([Sch13, Def. 3.6.134.], Cohomolgy in an ∞-topos) For two objects $X, A \in \mathcal{X}$ we define the cohomology set of X with coefficients in A by

$$H(X, A) \coloneqq \pi_0 \mathcal{X}(X, A)$$

where $\mathcal{X}(X, A)$ is the ∞-groupoid of morphisms between X and A. Generally, if A has a p-fold delooping for $p \in \mathbb{N}$ we define

$$H^p(X, A) \coloneqq \pi_0 \mathcal{X}(X, \mathbf{B}^p A)$$

The goal is now to write down a differential refinement of the cohomolgy in a cohesive ∞-topos. For this we need to define a version of De Rham cohomolgy that will later correspond to a truncated complex of differential form as used in the classical case.

1.2. Cohesive De-Rham complex.

Definition 1.7. ([Sch13, Def. 3.9.12]) Let $\mathcal{X}$ be a cohesive ∞-topos. For $X \in \mathcal{X}$ we define the cohesive de Rham homotopy type $\Pi_{\mathrm{dR}} X$ of X as the pushout

$$\begin{tikzcd} X \arrow[r] \arrow[d, "\eta"] & * \arrow[d] \\ \mathrm{Disc} \circ \Pi(X) \arrow[r] & \Pi_{\mathrm{dR}} X \end{tikzcd} \tag{1.8}$$

where η is the counit of the $\Pi \dashv \mathrm{Disc}$.

For a pointed object $* \to A$ in $\mathcal{H}$ we define the de Rham coefficient object $\flat_{\mathrm{dR}} A$ by the pullback

$$\begin{tikzcd} \flat_{\mathrm{dR}} A \arrow[r] \arrow[d] & \mathrm{Disc} \circ \Gamma(A) \arrow[d, "\epsilon"] \\ * \arrow[r] & A \end{tikzcd} \tag{1.9}$$

where ϵ is the counit of the adjunction $\mathrm{Disc} \dashv \Gamma$.

This also yields an adjunction

Proposition 1.10. *(*[Sch13, Prop. 3.9.13, Def 3.9.15]*) There is an adjunction*

$$(\Pi_{\mathrm{dR}} \dashv \flat_{\mathrm{dR}})\colon\ * /\mathcal{X} \underset{\flat_{\mathrm{dR}}}{\overset{\Pi_{\mathrm{dR}}}{\rightleftarrows}} \mathcal{X}$$

where $*/\mathcal{X}$ is the category of pointed objects in $\mathcal{X}$. We write

$$H_{\mathrm{dR}}(X, A) \coloneqq H(\Pi_{\mathrm{dR}} X, A) \cong H(X, \flat_{\mathrm{dR}} A)$$

for the de Rham cohomology of X with coefficients in A.

Now it is sensible to check wether this yield the well known de Rham cohomolgy in the case that $\mathcal{X}$ is a sheaf on a smooth manifold M.

Definition 1.11. Define the site of cartesian spaces CartSp as the category of cartesian spaces $\mathbb{R}^n$ for $n \in \mathbb{N}$ with continuous smooth functions between them.

Consider the category of simplicial presheaves $[\mathsf{CartSp}^{\mathrm{op}}, \mathsf{sSet}]$

Now we endow the category of simplical sets with the quillen model structure and localize at the maps

$$\check{C}(\{U_i \to U\}) \to U$$

where $\{U_i \to U\}$ is a covering family of U, $\check{C}$ denotes the simplicial manifold defined by the Čech nerve which gives a simplical presheaf by Yoneda embedding. Here U is considered as a simplicial presheaf by inserting the constant presheaf U in each degree, see [Dug01, Section 2]. We then obtain the category of ∞-sheaves $\mathcal{S}\mathrm{hv}_{\mathcal{M}}\mathrm{fd}$ valued in spaces, see [Lur09, Prop. 6.5.2.14.] and [Sch13, Thm. 2.2.15.].

Proposition 1.12. *([Sch13, Prop. 4.4.22., Def. 4.4.21.]) The simplical presheaf*

$$U(1)[n] \coloneqq [\cdots \to 0 \to \mathcal{C}^\infty(-, U(1)) \to 0 \to \cdots \to 0]$$

concentrated in degree n is a fibrant representative in $\mathcal{S}\mathrm{hv}_{\mathcal{M}\mathrm{fd}}$ of $\mathbf{B}^n U(1)$ under the Quillen equivalence of chain complexes with projective model structure and the Quillen model structure on simplicial sets, see see [Sch13, Prop. 2.2.31.].

Proposition 1.13. *([Sch13, Prop. 4.4.49.]) A fibrant representative in $\mathcal{S}\mathrm{hv}_{\mathcal{M}\mathrm{fd}}$ of the de Rham coefficient object $\flat_{\mathrm{dR}}\mathbf{B}^n U(1)$ is given by the truncated ordinary de Rham complex of smooth differential forms*

$$\flat_{\mathrm{dR}}\mathbf{B}^n U(1)_{\mathrm{chn}} \coloneqq \Psi[\Omega^1(-) \xrightarrow{\mathcal{D}_{\mathrm{dR}}} \Omega^2(-) \xrightarrow{\mathcal{D}_{\mathrm{dR}}} \cdots \xrightarrow{\mathcal{D}_{\mathrm{dR}}} \Omega^{n-1}(-) \xrightarrow{\mathcal{D}_{\mathrm{dR}}} \Omega^n_{\mathrm{cl}}(-)]$$

where Ψ denotes the Quillen equivalence between simplical sets with the Quillen model structure and the projective model structure on chain complexes, see [Sch13, Prop. 2.2.31.].

Proposition 1.14. *([Sch13, Prop. 4.4.50.]) There is are isomorphisms for $n \in \mathbb{N}$*

$$H^n_{\mathrm{dR}}(X, \mathbf{B}^n U(1)) \coloneqq \pi_0 \mathcal{S}\mathrm{hv}_{\mathcal{M}\mathrm{fd}}(M, \flat_{\mathrm{dR}}\mathbf{B}^n U(1)) \cong \begin{cases} H^n_{\mathrm{dR}}(M) & n \geq 2 \\ \Omega^1_{\mathrm{cl}}(M) & n = 1 \\ 0 & n = 0 \end{cases}$$

where a manifold considered as a constant sheaf.

1.3. **Differential cohomology.** Now we still need to define a characteristic map for the differential refinement we want to construct.

Definition 1.15. ([Sch13, Def. 3.9.29.]) For a group object $G \in \mathcal{X}$ in the cohesive ∞-topos $\mathcal{X}$ define the Maurer-Cartan Form

$$\theta \colon G \to \flat_{\mathrm{dR}}\mathbf{B}G$$

via the pullback diagrams:

$$\begin{tikzcd} G \arrow[r] \arrow[d, "\theta"] & * \arrow[d] \\ \flat_{\mathrm{dR}}\mathbf{B}G \arrow[r] \arrow[d] & \mathrm{Disc} \circ \Gamma(\mathbf{B}G) \arrow[d, "\epsilon"] \\ * \arrow[r] & \mathbf{B}G \end{tikzcd} \tag{1.16}$$

where, again ϵ is the counit of the adjunction.

Now the general definition of differential cohomology with coefficients in a braided ∞-group, that means the double delooping $\mathbf{B}^2 G$ exists, is given as follows:

Definition 1.17. ([Sch13, Def. 3.9.32., Rmk. 3.9.33.], General differential cohomology in a cohesive topos)

The differential cohomology with coefficients in $\mathbf{B}G$ is defined as the cohomology of the slice topos

$$\mathcal{X}/\flat_{\mathrm{dR}}\mathbf{B}^2 G.$$

A domain object in this is an object $X \in \mathcal{X}$ with a de Rham cocycle $F\colon X \to \flat_{\mathrm{dR}}\mathbf{B}^2 G$. An element in the corresponding cohomology group $\mathcal{X}/\flat_{\mathrm{dR}\mathbf{B}^2 G}((X,F), \mathrm{curv}_G := \theta\colon \mathbf{B}G \to \flat_{\mathrm{dR}}\mathbf{B}^2 G)$ is then given by a tranformation

$$\begin{tikzcd} X \arrow[rr, "g"] \arrow[dr, "F"'] & \arrow[d, phantom, "\Downarrow\nabla"] & \mathbf{B}G \arrow[dl, "\mathrm{curv}_G"] \\ & \flat_{\mathrm{dR}}\mathbf{B}^2 G & \end{tikzcd} \tag{1.18}$$

This is:

- a cocycle $g\colon X \to \mathbf{B}G$ for a G-principal bundle over X (in the cohesive sense [Sch13, Def 3.6.152.])
- an equivalence $g^*\mathrm{curv}_G \to F$ interpreted as a connection on the G-principal bundle.

Restricting to the special case that G is an Eilenberg-MacLane object, i.e. $\mathbf{B}G \cong \mathbf{B}^n A$ for a 0-truncated abelian group object A, see [Lur09, Def. 5.5.6.1.], we finally look at a more familiar diagram of a differential refinement.

Definition 1.19. For $X \in \mathcal{X}$ and $n \geq 1$ write

$$\mathcal{X}_{\mathrm{diff}}(X, \mathbf{B}^n A) := \mathcal{X}(X, \mathbf{B}^n A) \prod_{\mathcal{X}(X, \flat_{\mathrm{dR}}\mathbf{B}^n A)} H^{n+1}_{\mathrm{dR}}(X, A)$$

for the pullback of the diagram

$$\begin{tikzcd} \mathcal{X}_{\mathrm{diff}}(X, \mathbf{B}^n A) \arrow[r] \arrow[d] & H^{n+1}_{\mathrm{dR}}(X, A) \arrow[d] \\ \mathcal{X}(X, \mathbf{B}^n A) \arrow[r, "\mathrm{curv}_*"] & \mathcal{X}(X, \flat_{\mathrm{dR}}\mathbf{B}^n A) \end{tikzcd} \tag{1.20}$$

where $\mathrm{curv}\colon \mathbf{B}^n A \to \flat_{\mathrm{dR}}\mathbf{B}^{n+1} A$ is the Maurer-Cartan Form, in this case called the curvature characteristic morphism and the right vertical map is the unique, up to equivalence morphism from $H^{n+1}_{\mathrm{dR}}(X, A)$ seen as the 0-truncation of $\mathcal{X}(X, \flat_{\mathrm{dR}}\mathbf{B}^n A)$.

We call

$$H^n_{\mathrm{diff}}(X, A) := \pi_0 \mathcal{X}_{\mathrm{diff}}(X, \mathbf{B}^n A)$$

the degree-n differential cohomology of X with coefficients in A.

One (not me) can also see that in the case of $A = U(1) \cong K(\mathbb{Z}, 1)$ this definition gives the classical definition of Hopkins and Singer, see [HS05].

Theorem 1.21. *([Sch13, Thm. 4.4.87.]) For a smooth manifold M seen as a sheaf on* CartSp *we have that $H^n_{\mathrm{diff}}(M, U(1))$ is given by the subset of Deligne cocycles that picks for each de Rham cohomology class of M a curvature form representative.*

References

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